Black-Scholes From Scratch

Pablo Marchesi
Quantitative Analyst

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$$C_0 = S_0\Phi(d_1) - Ke^{-rT}\Phi(d_2)$$

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Introduction

In this presentation we will derive the ***Black-Scholes Formula*** from scratch, step by step, while connecting all the relevant topics in option pricing. Specifically, we will cover the following key items in option pricing theory:

- Options and Payoffs
- No-Arbitrage Pricing
- Risk-Neutral Measure
- Geometric Brownian Motion
- Change of Measure
- Derivation of the Black-Scholes Formula

Options and Payoffs

We will start by answering the fundamental question: ***What is an option?***

Financial Option

An option is a financial derivative that gives the holder the right, but not the obligation, to buy or sell an underlying asset at a predetermined price before or at a specific expiration date.

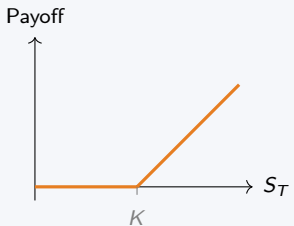
The predetermined price at which the underlying asset can be bought or sold is called the ***strike price*** K, and the specific expiration date is denoted by T. The price paid for acquiring the option is known as the ***premium***.

Options and Payoffs

There are two main types of options: call options and put options:

Call option

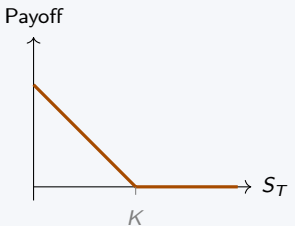
Right to ***buy*** the underlying at strike K.



Profits when the price ***rises***

Put option

Right to ***sell*** the underlying at strike K.



Profits when the price ***falls***

Options and Payoffs

The ***payoff*** of an option is the amount of money the holder receives at expiration, depending on the price of the underlying asset relative to the strike price.

For a call option with strike price K, the payoff at expiration, C_T, is:

$$C_T = \max(S_T - K, 0)$$

For a put option with strike price K, the payoff at expiration, P_T, is:

$$P_T = \max(K - S_T, 0)$$

Where S_T is the price of the underlying asset at expiration. In option pricing we use the subscript T to denote the moment of expiration.

Options and Payoffs

Options are a highly versatile instrument, with many applications in the financial industry. Some examples of option use cases are:

Selling	Banks generate income by selling (writing) options and collecting premiums.
Hedging	Protecting an existing investment from adverse price movements.
Speculation	Betting on the future price movements of the underlying asset.
Structuring	Creating complex financial products by combining multiple options and other derivatives.
Arbitrage	Exploiting price discrepancies between different markets or instruments.

No-Arbitrage Pricing

Option pricing theory is concerned with one key topic: ***What is the fair value of an option?*** Options must always have a non-zero price. Let us do a simple example to illustrate this concept:

Suppose the price of a call option, denoted as C_0, is equal to 0. Then I could buy the option for free and since:

$$\mathbb{P}(\text{option expires in the money}^1) > 0$$

I have a positive probability of making a profit with no initial investment (a risk-free profit):

$$\mathbb{P}(\text{Risk-Free Profit}) > 0$$

[1] In The Money (ITM) means the spot price is above the strike price for a call option or below the strike price for a put option.

No-Arbitrage Pricing

In summary, if options had a price of zero, I could buy all the options in the market for free and wait until some of them expire in the money.

The previous example is a demonstration of an ***arbitrage opportunity***, where one can make a risk-free profit.

Arbitrage Opportunity

An arbitrage opportunity is a strategy V where the initial investment is zero, $V_0 = 0$, and its terminal value V_T satisfies:

$$V_T \geq 0, \qquad \mathbb{P}(V_T > 0) > 0.$$

Arbitrage opportunities create infinite money paradoxes and we must ensure that our pricing models prevent them.

No-Arbitrage Pricing

Options must have a price greater than zero to prevent arbitrage. However, two questions arise:

Question 1

Is the price unique?

Can I have more than one price for the same option?

Question 2

How do we find it?

Is there a systematic method to determine the price (or prices) of an option?

Do prices depend on expectations?

Expecting Apple stock to rise could make a call option seem more valuable. However, as we will demonstrate, the no-arbitrage principle ensures that the option's price is ***independent of individual investors' expectations*** about the future price of the underlying asset.

No-Arbitrage Pricing

Let us consider two investors, A and B, with different beliefs about the future price of the underlying asset. Denote each investor's belief by a probability measure:

$$\mathbb{P}^A \text{ for Investor } A \text{ and } \mathbb{P}^B \text{ for Investor } B.$$

Probability Measure

A probability measure $\mathbb{P}$ is a function that assigns probabilities to different possible outcomes of a random variable. More formally:

$$\mathbb{P} : \mathcal{F} \to [0, 1],$$

where $\mathcal{F}$ is a sigma-algebra of subsets of Ω, the set of all possible outcomes of the random variable.

No-Arbitrage Pricing

In our context, we can think of a probability measure as a way to assign probabilities to different possible future prices of the underlying asset.

$$\mathbb{P} : \textit{Future Scenarios} \longrightarrow [0, 1]$$

For example, investor A might think that:

$$\mathbb{P}^A(S_T > 100) = 40\%$$

Similarly, investor B might have a different belief:

$$\mathbb{P}^B(S_T > 100) = 70\%$$

No-Arbitrage Pricing

Investors can estimate the asset's expected future price using their own probability measure. For example, investor A might believe:

$$\mathbb{E}^A[S_T] = 80$$

While for investor B:

$$\mathbb{E}^B[S_T] = 100$$

The expected value is a probability-weighted average of all possible outcomes.

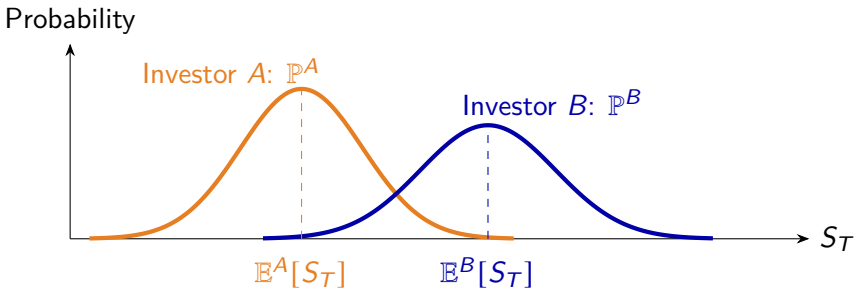
Expected Value

If X has probability density function f_X under $\mathbb{P}$, its expected value is:

$$\mathbb{E}^{\mathbb{P}}[X] = \int_{-\infty}^{\infty} x \cdot f_X(x)\, dx,$$

No-Arbitrage Pricing

In summary, each investor has their own market view represented by their probability measure and expected value.



No-Arbitrage Pricing

While it is totally normal to have different beliefs and expectations about market prices, let us now consider the following problem:

What is the price of a contract that pays 100 ***at time*** T ***if a given stock price*** S_T ***exceeds*** 50 ***at that time?***

If we formalize this mathematically:

- Denote V_0 as the price of the contract at the current time.
- Denote S_T as the stock price at time T.
- Denote K as the threshold stock price, with $K = 50$.
- Denote V_T as the payoff of the contract at time T, which is 100 if $S_T > K$ and 0 otherwise.

No-Arbitrage Pricing

Going back to our two investors, A and B, they might have different valuations for the same contract based on their individual beliefs:

$$\begin{cases} V_0^A = 40, & \text{given that } \mathbb{E}^A[S_T] = 75, \\ V_0^B = 80, & \text{given that } \mathbb{E}^B[S_T] = 110. \end{cases}$$

Which, intuitively, may seem reasonable:

- Investor A has a lower expected stock price and thus values the contract lower.
- Investor B has a higher expected stock price and values the contract higher.

No-Arbitrage Pricing

Now consider a smart investor C whose beliefs are denoted by $\mathbb{P}^{\mathbb{Q}}$. Investor C sets up the following strategy to exploit the discrepancy between investors A and B:

1. Creates a portfolio of a long and a short[2] position in the contract.
2. Buys the contract from Investor A at the lower price of $V_0^A = 40$.
3. Shorts the contract from Investor B at the higher price of $V_0^B = 80$.
4. Makes a profit of $V_0^B - V_0^A = 80 - 40 = 40$.
5. At T, the long and short positions offset each other, regardless of S_T.

This strategy ensures a ***risk-free profit*** for Investor C, and constitutes an arbitrage opportunity.

[2] A short position collects the premium upfront but obligates the investor to deliver the contract payoff at maturity.

No-Arbitrage Pricing

To prevent this arbitrage opportunity, the following condition must be met:

$$V_0^A = V_0^B = V_0$$

Which leads to the following no-arbitrage condition:

No-Arbitrage Condition

In a market free of arbitrage opportunities, the price of a contract must be the same for all investors, regardless of their individual beliefs.

However, if $V_0^A = V_0^B$, what happens with the individual beliefs of investors A and B, $\mathbb{P}^A$ and $\mathbb{P}^B$?

Risk-Neutral Measure

The uniqueness of prices implies that all investors must agree on the probability measure used to value the contract, called the ***risk-neutral measure*** or ***pricing measure***, denoted $\mathbb{P}^{\mathbb{Q}}$.

More formally, we can define two different probability measures in the market:

Physical measure, $\mathbb{P}$

- Real-world distribution of the asset's future price.
- Estimated from historical data and market information.
- Produces arbitrage opportunities if used to price derivatives.

Use: Predicting Prices

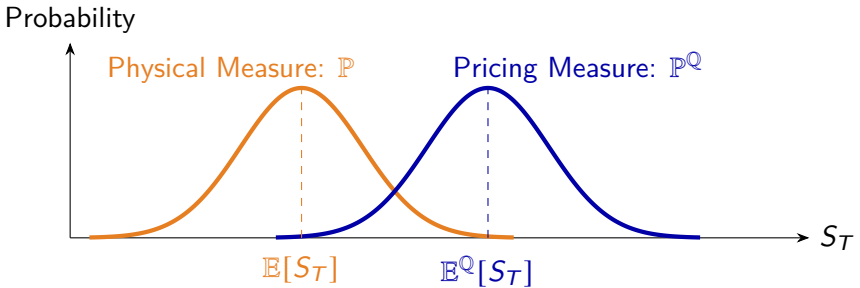
Pricing measure, $\mathbb{P}^{\mathbb{Q}}$

- Risk-neutral distribution of the asset's future price.
- Prices derivatives consistently across all investors.
- Eliminates arbitrage opportunities when used to price derivatives.

Use: Valuing Derivatives

Risk-Neutral Measure

We use the physical measure $\mathbb{P}$ to predict stock prices, while the pricing measure $\mathbb{P}^{\mathbb{Q}}$ is used to value financial derivatives such as options.



While $\mathbb{P}^{\mathbb{Q}}$ is easy to obtain (from derivative prices), the physical measure $\mathbb{P}$ is more challenging to estimate, as it would imply that we can predict the actual future prices of the asset.

Risk-Neutral Measure

Equipped with the risk-neutral measure $\mathbb{P}^{\mathbb{Q}}$, we can now price any financial derivative we wish by using the following formula:

Derivatives Pricing Formula

The value of a financial derivative at time 0, denoted as V_0, is given by

$$V_0 = e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[V_T\right]$$

Where V_T is the payoff at time T, and r is the risk-free interest rate.

In the case of a call option with strike price K, the value, C_0, at time 0 is given by:

$$C_0 = e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[C_T\right] = e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[\max(S_T - K, 0)\right]$$

Geometric Brownian Motion

The next question we ask ourselves is: how can we model future asset prices, S_T, under both the physical measure $\mathbb{P}$ and the risk-neutral measure $\mathbb{P}^{\mathbb{Q}}$?

Let us start with the physical measure $\mathbb{P}$.

Geometric Brownian Motion under the Physical Measure

An asset price (such as a stock) under $\mathbb{P}$ can be modeled using a stochastic differential equation as follows:

$$\frac{dS_t}{S_t} = \mu\, dt + \sigma\, dW_t^{\mathbb{P}},$$

where μ is the drift, σ is the volatility, and $W_t^{\mathbb{P}}$ is a standard Brownian motion under the physical measure $\mathbb{P}$.

Geometric Brownian Motion

At first, this equation might seem a bit abstract, but it essentially states the following:

$$\underbrace{\frac{dS_t}{S_t}}_{\text{Change in Stock Price}} = \underbrace{\mu\, dt}_{\text{Deterministic Component}} + \underbrace{\sigma\, dW_t^{\mathbb{P}}}_{\text{Random Component}},$$

Some key takeaways from this equation are:

- $\mu\, dt$ represents the expected return of the stock.
- $dW_t^{\mathbb{P}}$ represents the increment of a Brownian Motion (normally distributed).
- $\sigma\, dW_t^{\mathbb{P}}$ represents the uncertainty in the stock price.

Geometric Brownian Motion

Solving this equation[3] we get the following solution, where S_0 is the current stock price:

$$S_t = S_0 \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t^{\mathbb{P}}\right]$$

We can also express $W_t^{\mathbb{P}}$ in terms of a normal random variable as follows:

$$W_t^{\mathbb{P}} = \sqrt{t}Z, \quad Z \sim \mathcal{N}(0,1)$$

So we can rewrite the solution for S_t as:

$$S_t = S_0 \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma\sqrt{t}Z\right], \quad Z \sim \mathcal{N}(0,1)$$

[3] Using Itô's Lemma

Geometric Brownian Motion

We are usually interested in the distribution of the stock price at a future time T, therefore:

$$S_T = S_0 \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}Z\right], \quad Z \sim \mathcal{N}(0,1)$$

And, since this is a function of a normally distributed random variable, S_T is lognormally[4] distributed, and has the following expectation:

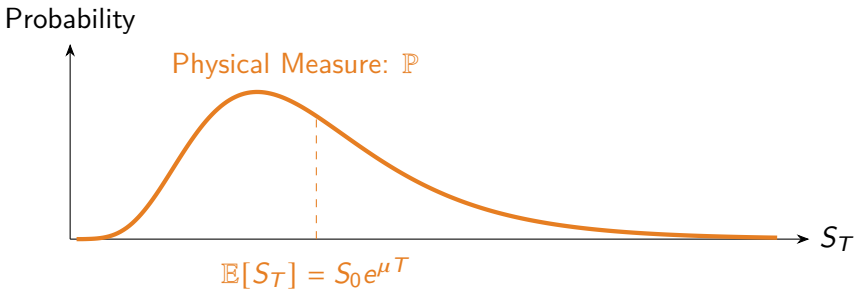
$$\mathbb{E}[S_T] = S_0 e^{\mu T}$$

Clearly, the μ parameter is very hard to estimate under $\mathbb{P}$, as it would imply that we have precise knowledge of the future value of the stock price.

[4] In the GBM, stock returns are normally distributed and prices are lognormal.

Geometric Brownian Motion

Under the physical measure $\mathbb{P}$, the stock price at time T follows a lognormal distribution[5].



[5]Note that, since stock prices can never be negative, the lognormal distribution is a more appropriate model for stock prices than the normal distribution.

Change of Measure

We now ask ourselves how the stock price behaves under $\mathbb{P}^{\mathbb{Q}}$. Since we want to keep the lognormal form of the stock price, we require that:

$$\mathbb{E}^{\mathbb{Q}}[S_T] = S_0 e^{\hat{\mu} T}$$

Where $\hat{\mu}$ is the drift of the stock price under $\mathbb{P}^{\mathbb{Q}}$. The key question in risk-neutral pricing is: what should the value of $\hat{\mu}$ be? To answer this question we need to understand the concept of a martingale:

Intuitive Definition of Martingales

A stochastic process X_t is called a martingale with respect to a probability measure $\mathbb{P}$ if the following condition is met:

$$\mathbb{E}[X_t] = X_0$$

Change of Measure

Some key takeaways about martingales:

- Represent the notion of a *fair game*.
- Imply that our best prediction for a future value of the process is its current value.
- Rule out arbitrage opportunities in financial markets.
- Are central to the concept of risk-neutral pricing.

In risk-neutral pricing, discounted[6] asset prices are martingales under the risk-neutral measure $\mathbb{Q}$. Mathematically, this can be expressed as:

$$\mathbb{E}^{\mathbb{Q}}\left[\widehat{S}_T\right] = \mathbb{E}^{\mathbb{Q}}\left[e^{-rT}S_T\right] = \widehat{S}_0 = S_0$$

[6] Where $\widehat{S}_t = e^{-rt}S_t$ is the discounted stock price at time t. Note that $\widehat{S}_0 = S_0$.

Change of Measure

Therefore, if under the risk-neutral measure $\mathbb{Q}$, discounted asset prices are martingales, it follows that:

$$\mathbb{E}^{\mathbb{Q}}\left[e^{-rT}S_T\right] = e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[S_T\right] = e^{-rT}S_0e^{\hat{\mu}T} = S_0e^{(\hat{\mu}-r)T} = S_0$$

This implies that the drift of the stock price under the risk-neutral measure is the risk-free rate r, in other words:

$$\boxed{\hat{\mu} = r}$$

Consequently, the expectation of future stock prices under the risk-neutral measure grows at the risk-free rate:

$$\mathbb{E}^{\mathbb{Q}}[S_T] = S_0e^{rT}$$

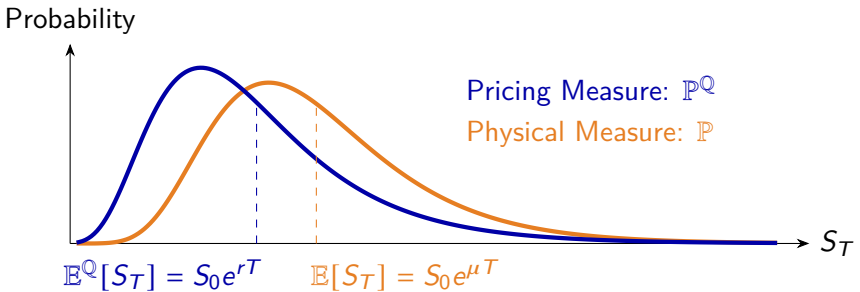
Change of Measure

Some key implications of the change to the risk-neutral measure are:

- All the assets grow on average at the risk-free rate.
- Investors are indifferent to risk, or risk-neutral, as every asset grows at the same rate.
- Discounted asset prices do not go up or down on average. No stock is expected to outperform the others.
- Option prices are unique as every investor uses the same risk-neutral measure for pricing and expects the asset prices to grow at the risk-free rate.
- The market is a *fair game* and there are no arbitrage opportunities.

Change of Measure

Therefore, we will have two lognormal distributions for the stock price at time T. One under the physical measure $\mathbb{P}$ and another under the risk-neutral measure $\mathbb{P}^{\mathbb{Q}}$.



Derivation of the Black-Scholes Formula

We now have all the ingredients to finally derive the Black-Scholes formula. Let us consider a call option with strike K and maturity T. Its price at time 0 is given by:

$$C_0 = e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[\max(S_T - K, 0)\right]$$

Under $\mathbb{P}^{\mathbb{Q}}$, the stock price is:

$$S_T = S_0 \exp\left[\left(r - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}Z\right], \quad Z \sim \mathcal{N}(0,1)$$

so the expectation is an integral against the standard normal density $\varphi(z) = \frac{1}{\sqrt{2\pi}}e^{-z^2/2}$:

$$C_0 = e^{-rT}\int_{-\infty}^{\infty} \max(S_T - K, 0)\frac{1}{\sqrt{2\pi}}e^{-\frac{z^2}{2}}\,dz$$

Derivation of the Black-Scholes Formula

The payoff is positive only when $S_T > K$. Consequently, by taking logarithms, we have:

$$S_T > K \iff \left(r - \tfrac{1}{2}\sigma^2\right) T + \sigma\sqrt{T}z > \ln\frac{K}{S_0} \iff z > -d_2$$

where

$$d_2 = \frac{\ln\left(S_0/K\right) + \left(r - \tfrac{1}{2}\sigma^2\right) T}{\sigma\sqrt{T}}$$

Therefore, the integral only needs to run over $z > -d_2$ and the positive part can be dropped:

$$C_0 = e^{-rT} \int_{-d_2}^{\infty} \left(S_0 e^{(r-\frac{1}{2}\sigma^2)T + \sigma\sqrt{T}z} - K\right) \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}\, dz$$

Derivation of the Black-Scholes Formula

Splitting the integral:

$$C_0 = e^{-rT}\left[S_0 \underbrace{\int_{-d_2}^{\infty} e^{(r-\frac{1}{2}\sigma^2)T+\sigma\sqrt{T}z}\,\varphi(z)\,dz}_{I_1} - K \underbrace{\int_{-d_2}^{\infty} \varphi(z)\,dz}_{I_2}\right]$$

The second integral, I_2, is the probability that the option finishes in the money. Using the symmetry of the normal distribution, where Φ is the standard normal CDF:

$$I_2 = \mathbb{P}^{\mathbb{Q}}(Z > -d_2) = \Phi(d_2)$$

Derivation of the Black-Scholes Formula

For the first integral, I_1, we pull the constant e^{rT} out and combine the exponents by completing the square:

$$I_1 = e^{rT} \int_{-d_2}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\sigma^2 T + \sigma\sqrt{T}z - \frac{1}{2}z^2}\, dz$$

$$-\frac{1}{2}\sigma^2 T + \sigma\sqrt{T}z - \frac{1}{2}z^2 = -\frac{1}{2}\left(z - \sigma\sqrt{T}\right)^2$$

so that

$$I_1 = e^{rT} \int_{-d_2}^{\infty} \varphi\left(z - \sigma\sqrt{T}\right) dz$$

Derivation of the Black-Scholes Formula

Substituting $u = z - \sigma\sqrt{T}$, the lower limit of the integral becomes $-d_2 - \sigma\sqrt{T}$:

$$I_1 = e^{rT} \int_{-d_2-\sigma\sqrt{T}}^{\infty} \varphi(u)\, du = e^{rT}\, \Phi\left(d_2 + \sigma\sqrt{T}\right)$$

Defining

$$d_1 = d_2 + \sigma\sqrt{T} = \frac{\ln\left(S_0/K\right) + \left(r + \frac{1}{2}\sigma^2\right) T}{\sigma\sqrt{T}}$$

we obtain

$$\boxed{I_1 = e^{rT}\, \Phi(d_1)}$$

Derivation of the Black-Scholes Formula

Substituting I_1 and I_2 in the expression for C_0:

$$C_0 = e^{-rT}\left(S_0 e^{rT}\,\Phi(d_1) - K\,\Phi(d_2)\right)$$

Distributing the discount factor, we arrive at the ***Black-Scholes Formula***:

$$\boxed{C_0 = S_0\,\Phi(d_1) - Ke^{-rT}\,\Phi(d_2)}$$

with

$$d_1 = \frac{\ln\left(S_0/K\right) + \left(r + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}$$

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Thank you!